When Does a Moored Tanker Fishtail?

Pitchfork and Hopf Bifurcations of a Single-Point-Moored Tanker in Steady Current

Course Project Report

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1 Introduction

A single-point mooring holds a tanker at the bow by one hawser running to a buoy and leaves it free to weathervane. The ship does not always settle behind the buoy. Under some conditions it swings from side to side and keeps swinging even in a perfectly steady current. This motion is called fishtailing, and it matters because the hawser tension rises and falls with every swing.

If the bow moves off the current, the Munk moment tries to turn the hull further broadside, while the hawser tension and the side force on the hull pull it back. Depending on where the hawser is tied, how long it is and how large the friction drag is, the ship trails straight, rests at a steady angle or fishtails. Simos [1] built a hydrodynamic model for this problem and tested it on a 1:90 tanker model, and fishtailing has also been seen in flume tests [3]. Because the current is steady, the swing is a self-excited limit cycle, so the problem is a natural fit for the bifurcation theory of our course text [2]. We rebuilt Simos's model and studied six cases: the attachment point, friction, hawser length, a side wind, a hold-back tug and a stern thruster.

2 Methodology

Ship. We use the Vidal de Negreiros, a 320 m VLCC, in the fully loaded condition. Its particulars, taken from Simos [1], are listed in Table 1.

Table 1: Particulars of the Vidal de Negreiros (full scale, fully loaded).

Length L (m)	320	Mass M (kg)	3.219×10^8
Beam B (m)	54.5	Yaw inertia I_z (kg m^2)	2.06×10^{12}
Draft T (m)	21.47	Added mass, surge M_{11} (kg)	1.848×10^7
Block coefficient C_B	0.83	Added mass, sway M_{22} (kg)	2.725×10^8
CG forward of midship x_{CG} (m)	9.81	Added yaw inertia M_{66} (kg m^2)	1.579×10^{12}
Cross-flow drag coefficient C_Y	0.84	Sway–yaw coupling M_{26} (kg m)	2.209×10^9

Model. The hawser is a rigid link of length L_H, tied a distance x_A forward of midship (Fig. 1). With the buoy at the origin and the X axis pointing upstream, the attachment point and midship are at

$$\mathbf{A} = L_H(-\cos\gamma,\ \sin\gamma), \qquad \mathbf{O} = \mathbf{A} - x_A(\cos\psi,\ \sin\psi), \tag{1}$$

so the hawser angle γ and the heading ψ fix the position of the ship, and the state is $(\gamma, \psi, \dot\gamma, \dot\psi)$. Differentiating $\mathbf{O}$ and adding the current U gives the velocities relative to the water,

$$u = U\cos\psi + L_H\dot\gamma\sin(\gamma+\psi), \qquad v = -U\sin\psi + L_H\dot\gamma\cos(\gamma+\psi) - x_A\dot\psi, \qquad r = \dot\psi. \tag{2}$$

The hawser pulls the attachment point towards the buoy with force F_H, which gives $F_X = F_H\cos(\gamma+\psi)$, $F_Y = -F_H\sin(\gamma+\psi)$ and a yaw moment $-x_A F_H\sin(\gamma+\psi)$. With the water loads X_R, Y_R, N_R from Simos's model, $m_{26} = Mx_{CG} + M_{26}$ and $I_{66} = I_Z + M_{66}$ (yaw inertia about midship), the manoeuvring equations in surge, sway and yaw are

$$\begin{aligned} (M+M_{11})\dot u - (M+M_{22})vr - m_{26}r^2 &= X_R + F_H\cos(\gamma+\psi),\\ (M+M_{22})\dot v + (M+M_{11})ur + m_{26}\dot r &= Y_R - F_H\sin(\gamma+\psi),\\ I_{66}\dot r + m_{26}(\dot v + ur) &= N_R - x_A F_H\sin(\gamma+\psi). \end{aligned} \tag{3}$$

Since the hawser length is fixed, F_H is an unknown. Writing $\dot u = L_H s\,\ddot\gamma + a_u$ and $\dot v = L_H c\,\ddot\gamma - x_A\ddot\psi + a_v$, with $s = \sin(\gamma+\psi)$, $c = \cos(\gamma+\psi)$ and a_u, a_v the known remainders, turns the three equations into a linear system that we solve at every time step:

$$\begin{bmatrix} (M+M_{11})L_H s & 0 & -c\\ (M+M_{22})L_H c & m_{26}-(M+M_{22})x_A & s\\ m_{26}L_H c & I_{66}-m_{26}x_A & x_A s \end{bmatrix} \begin{bmatrix} \ddot\gamma\\ \ddot\psi\\ F_H \end{bmatrix} = \begin{bmatrix} X_R + (M+M_{22})vr + m_{26}r^2 - (M+M_{11})a_u\\ Y_R - (M+M_{11})ur - (M+M_{22})a_v\\ N_R - m_{26}(a_v+ur) \end{bmatrix}. \tag{4}$$

Current speed. Time is measured as $t^* = Ut/L$, the time the current takes to pass one ship length. In these units the current speed survives only through the skin-friction coefficient C_F, which falls as the current speeds up, so we use C_F as the speed parameter. It is about 0.02–0.05 for the 1:90 tank model

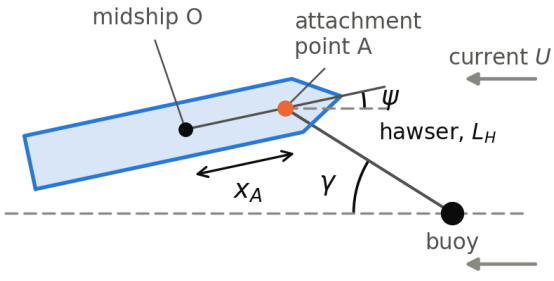


Figure 1: Plan view of the mooring. γ and ψ are positive to port; the bow is at $x_A/L = 0.5$.

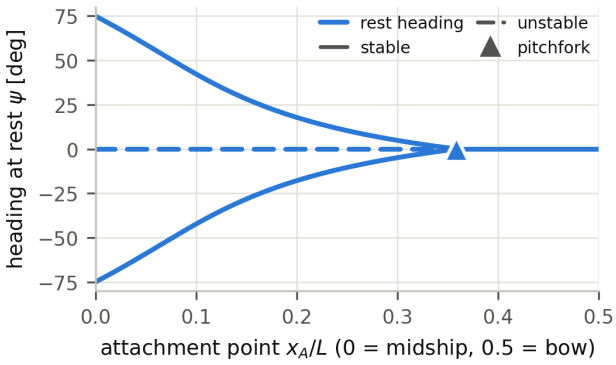


Figure 2: Rest heading against x_A/L ($L_H/L = 0.5$, $C_F = 0.05$).

and about 0.010 for the real ship in a 1 m/s current. When the ship trails straight, the hawser tension equals the friction drag, $F_{H0} = C_F \frac{1}{2}\rho U^2 LT$, and all tug, wind and thruster forces are given as multiples of F_{H0}.

Stability and limit cycles. A rest position is stable when all four eigenvalues $\lambda = \sigma \pm i\omega$ of the linearised system have negative real parts. A real eigenvalue crossing zero signals a pitchfork or a fold, and a complex pair crossing the imaginary axis signals a Hopf bifurcation. Limit cycles were found by simulation and, near a Hopf point, by a shooting method, with Floquet multipliers for their stability. Our Hopf boundary agrees with Simos's published curve to within +0.5 % on average (Fig. 4a).

Initial conditions. Each run starts from straight trailing with a small nudge, $\gamma = 3°$, $\psi = 1°$ and zero rates. In parameter sweeps each run starts from where the previous one ended, and sweeping in both directions reveals any hysteresis. In the control studies the runs start from the fully developed fishtail, with the control switched on at $t^* = 0$, and in the wind study they start with the ship resting at an angle to port.

3 Results

3.1 Attachment point: pitchfork

- **Set-up:** x_A/L varied (0 = midship, 0.5 = bow); $L_H/L = 0.5$ and a high friction $C_F = 0.05$ held fixed, so the ship never fishtails here.
- At rest the hawser force drops out of the equations, leaving

$$N_R(\psi) - x_A\, Y_R(\psi) = 0,$$

  so the rest heading depends only on x_A/L: the water's turning moment about the hawser point must vanish.
- Forward of $x_A/L = 0.359$ the hull's side force acts like a weathervane tail and the ship trails straight. Aft of it the Munk moment wins and the ship settles at a steady angle to port or starboard, e.g. $\psi = 10.3°$ at $x_A/L = 0.25$: a pitchfork (Fig. 2).
- A real eigenvalue (a slow sideways drift) crosses zero. Just past the pitchfork, straight trailing has $+0.034$ and the angled positions -0.034, an exchange of stability.
- It is supercritical: the new branches are stable and grow from zero. They grow linearly (log–log slope 1.00), not as a square root, because cross-flow drag varies as $\psi|\psi|$, giving $\dot{\psi} = r\psi - q\psi|\psi|$ in place of Strogatz's $\dot{x} = rx - x^3$.

3.2 Friction: Hopf bifurcation

- **Set-up:** C_F varied (lower = faster current); hawser at the bow ($x_A/L = 0.5$) and $L_H/L = 0.5$.
- The ship on its hawser behaves like a pendulum whose restoring force is the tension F_{H0}, which is proportional to C_F.
- A complex pair $\lambda = \sigma \pm i\omega$ crosses the imaginary axis ($\sigma = 0$, $\omega \neq 0$) at $C_F = 0.0425$, with $\omega = 0.738\, U/L$, a period of $8.5\, L/U$. This is a Hopf bifurcation.

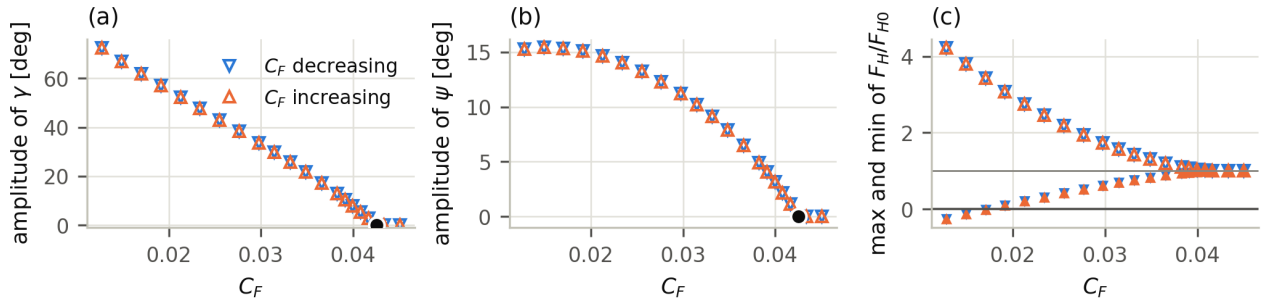


Figure 3: Effect of friction ($x_A/L = L_H/L = 0.5$): swing amplitude of (a) γ and (b) ψ, and (c) the largest (open) and smallest (filled) hawser force. • marks the Hopf point.

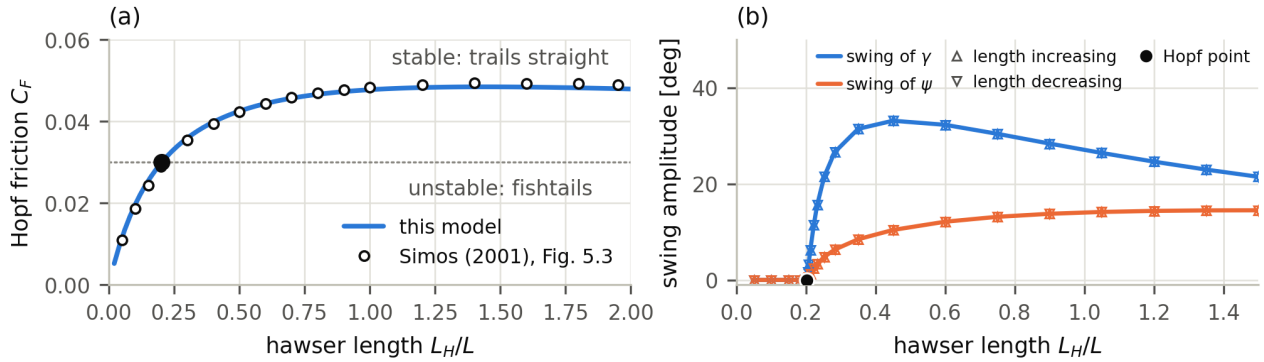


Figure 4: Effect of hawser length. (a) Hopf boundary of straight trailing, compared with Simos [1]; the dotted line marks $C_F = 0.03$. (b) Swing amplitude at $C_F = 0.03$ for increasing and decreasing hawser length.

- Below it the swing grows and levels off at a fixed size: $\gamma = \pm 33°$, $\psi = \pm 11°$ at $C_F = 0.03$, and $\gamma = \pm 72°$ at $C_F = 0.013$ (Fig. 3).
- Sweeps down and up give the same curve, so there is no hysteresis and the Hopf bifurcation is supercritical. Floquet multipliers confirm that the cycles are stable.
- The amplitude a grows in proportion to the distance past the Hopf point (slope 0.98), since averaging with quadratic drag gives $\dot{a} = \mu a - c a^2$ and hence $a^* = \mu/c$.
- Below $C_F \approx 0.017$ the hawser tension would turn negative for part of each swing, which a real rope cannot do.

3.3 Hawser length: Hopf bifurcation

- **Set-up:** L_H/L varied; hawser at the bow and $C_F = 0.03$.
- A short hawser is a stiff, well-damped pendulum. The Hopf condition $\sigma(L_H) = 0$ is met at $L_H/L = 0.202$, a hawser of about 65 m, and any longer hawser fishtails.
- Over all lengths (Fig. 4a) the boundary rises from $C_F = 0.0298$ at $L_H/L = 0.2$ to 0.0478 at 1.0, so shorter hawsers are safer. The real ship, at $C_F \approx 0.010$, fishtails for all but the shortest hawsers.
- Sweeps of increasing and decreasing length give the same swing, to within 0.05°. It grows from zero at the Hopf point and every cycle is stable, so this Hopf bifurcation is also supercritical (Fig. 4b).
- The swing angle γ peaks at about $\pm 33°$ near $L_H/L = 0.5$ and then falls slowly, but the bow's sideways sweep keeps growing, from ± 87 m at $L_H/L = 0.5$ to ± 176 m at 1.5, while the heading swing levels off near $\pm 14.5°$.

3.4 Wind: imperfect pitchfork

- **Set-up:** wind force F_w/F_{H0} and direction varied; $x_A/L = 0.25$ (aft of the pitchfork), $L_H/L = 0.5$ and $C_F = 0.05$. The wind is a constant force acting at midship.
- A side wind breaks the mirror symmetry. The pitchfork splits into one smooth branch and one that ends at a fold (Fig. 5a). The reduced equation is Strogatz's imperfect pitchfork with $\psi|\psi|$ in place of

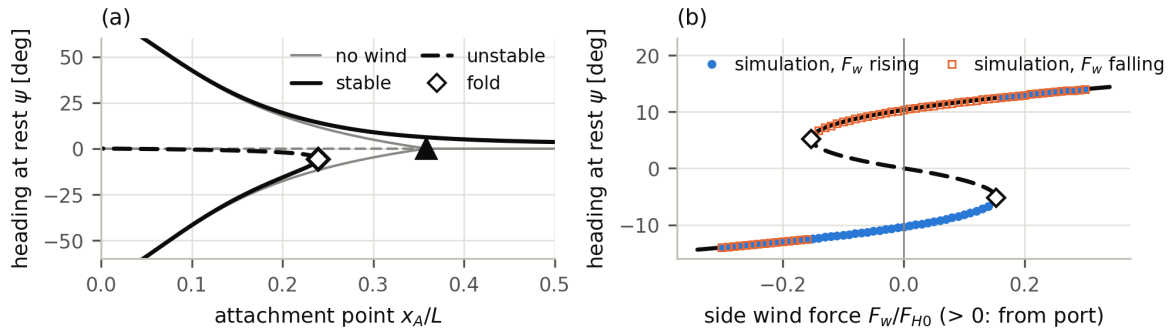


Figure 5: Effect of a side wind. (a) Imperfect pitchfork for a beam wind of 0.20 F_{H0} from port, over the no-wind diagram. (b) Hysteresis at $x_A/L = 0.25$; simulated sweeps follow the stable branches and jump just past the folds.

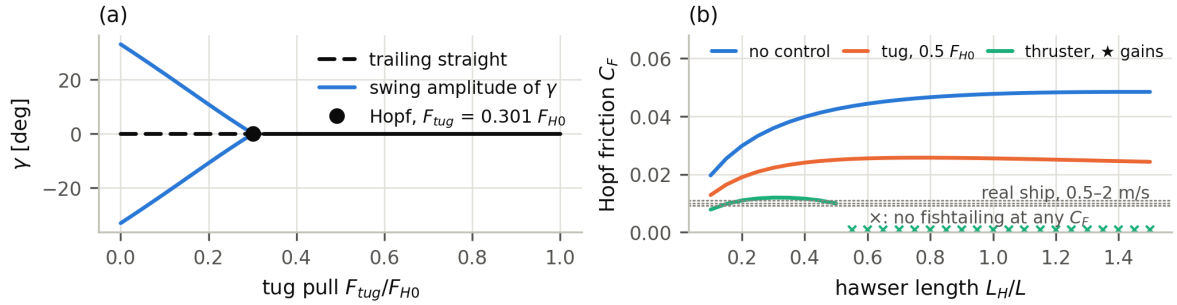


Figure 6: (a) Hold-back tug at $C_F = 0.03$: the swing shrinks to a Hopf point. (b) Hopf boundary with and without control at $x_A/L = 0.5$; the thruster uses the gains ★ of Fig. 7.

x^3,

$$\dot{\psi} = h + r\psi - q\psi|\psi|, \qquad \text{folds at } h = \pm\frac{r^2}{4q},$$

where h is the side part of the wind force.

- The ship keeps its side until the wind passes the fold, at $\pm 0.153\, F_{H0}$ for a beam wind, and then swings across. It does not swing back when the wind drops: hysteresis (Fig. 5b).
- At the fold a stable and an unstable rest position meet with a zero eigenvalue and vanish (saddle-node). Other directions need more wind, and wind from ahead or astern keeps the pitchfork perfect.
- The ship ends up on the upwind side. The wind acts aft of the hawser point and turns the bow into the wind, and the hull's lift in the current then carries the ship that way.

3.5 Control by tug

- **Set-up:** tug pull F_{tug}/F_{H0} and C_F varied; hawser at the bow and $L_H/L = 0.5$.
- A hold-back tug at the stern pulls steadily downstream. It raises the hawser tension to $F_{H0} + F_{tug}$, and once the bow swings off, the sideways part of its pull, about $F_{tug}\psi$, acts like a weathervane tail.
- At $C_F = 0.03$ the swing shrinks linearly with the pull and disappears at a Hopf point, $F_{tug} = 0.301\, F_{H0}$ (Fig. 6a).
- A faster current needs much more pull: $0.66\, F_{H0}$ at $C_F = 0.022$ and $2.24\, F_{H0}$ at $C_F = 0.010$, the real ship at about 1 m/s.
- A pull of $0.5\, F_{H0}$ lowers the Hopf friction at $L_H/L = 0.5$ from 0.0425 to 0.025 (Fig. 6b).

3.6 Control by stern thruster

- **Set-up:** gains k_d, k_p and thrust limit varied; $x_A/L = L_H/L = 0.5$ and $C_F = 0.03$.
- The ship's own stern thruster, at $x_T = -L/2$, follows a PD law on the heading:

$$N_c^* = -k_p\psi - k_d r^*, \qquad Y_T = \frac{N_c}{x_T}, \qquad |Y_T| \le \text{limit} \times F_{H0},$$

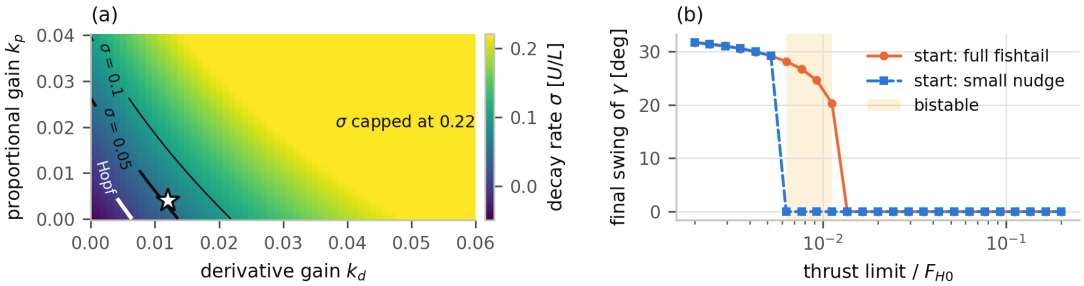


Figure 7: Stern thruster at $C_F = 0.03$. (a) Decay rate over the gain plane; ★ marks the least-thrust gains. (b) Swing left after 600 t^* against the thrust limit; the shaded band is bistable.

where $N_c^* = N_c/(\frac{1}{2}\rho U^2 L^2 T)$ and $r^* = rL/U$.

- k_d adds the damping that the Hopf instability lacks: $k_d = 0.0064$ alone stops the swing, or $k_p = 0.0131$ alone.
- The gains needing the least thrust for a decay rate $\sigma = -\max \mathrm{Re}\,\lambda \geq 0.05\,U/L$ are $k_p = 0.004$, $k_d = 0.012$, with a peak thrust of $0.082\,F_{H0}$ (Fig. 7a).
- One eigenvalue, $-0.221\,U/L$, a slow sideways drift that heading control cannot see, never moves, so $\sigma \leq 0.22$.
- A thrust limit does not change the eigenvalues but matters for large swings. Between 0.0063 and $0.0113\,F_{H0}$ the system is bistable: the full fishtail survives while a small nudge dies out (Fig. 7b).
- With these gains the Hopf friction drops to 0.0099 at $L_H/L = 0.5$, and for $L_H/L \geq 0.55$ fishtailing is gone at every C_F (Fig. 6b).

4 Conclusions and limitations

Each behaviour of the moored ship matches a standard bifurcation (Table 2), and because the limiting force is quadratic drag the branches grow linearly.

Table 2: Bifurcations found in the six cases.

Case	Parameter	Bifurcation	Threshold
Attachment point	x_A/L	Supercritical pitchfork	0.359
Friction	C_F	Supercritical Hopf	0.0425
Hawser length	L_H/L	Supercritical Hopf	0.202 ($C_F = 0.03$)
Wind	F_w/F_{H0}	Imperfect pitchfork, hysteresis	folds at ± 0.153 (beam wind)
Tug	F_{tug}/F_{H0}	Hopf	0.301 ($C_F = 0.03$)
Thruster	k_d, k_p	Hopf, bistable with a thrust limit	$k_d = 0.0064$ or $k_p = 0.0131$

- The hawser is rigid. In a separate run with a measured nylon rope law [4] the boundaries moved by less than 5 %, but at 1 m/s the rope was slack 74 % of the time and snatch loads reached $14\,F_{H0}$.
- Flume tests have shown a subcritical Hopf bifurcation [3], unlike our model, so the type may depend on the set-up.
- The wind is a constant force, tug and thruster sizes are ratios as we found no cited VLCC values, and the tug as a cure for fishtailing is a result of our model only.
- An interactive version of these results is available at `https://aaravpatel007.github.io/Moored-Tanker-Fishtailing/interactive/`.

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[4] H. Huang et al., Investigation into fishtailing effect of oil tankers moored at pile-founded column SPM systems, *J. Mar. Sci. Eng.* 14, 770, 2026.